

TEST AUTOMATION PROJECT REPORT

Ixigo Flight Booking Automation

Selenium WebDriver, pytest, Page Object Model, pytest-html and Allure

Application under test: ixigo flight booking web journey

Automation stack: Python, Selenium WebDriver, pytest, Page Object Model

Execution browser: Microsoft Edge, non-headless mode

Report date: 22 May 2026

Result summary: 24 Selenium checks documented from the combined flow and the full end-to-end booking flow

1. Synopsis

The project automates the main ixigo flight-booking path using Selenium WebDriver. The suite starts from the flight-search page, validates core search behaviour, applies practical result-page actions such as filtering and sorting, and then continues through booking, traveller details, add-ons, seat selection and the payment page. The goal is not only to click through screens, but to prove that the most important user journey still behaves correctly when handled by an automated test framework.

The work is organised as a Selenium pytest framework with page objects, reusable utilities, CSV-based test data, screenshots and HTML/Allure reporting. This keeps the test code readable and makes the result easier to review during a capstone-style evaluation.

2. Project Description

The application under test is the ixigo flight booking website. A normal user journey contains several moving parts: source and destination selection, calendar interaction, result loading, pop-up handling, price sorting, booking navigation, traveller form entry, add-ons and payment routing. These are exactly the areas where manual testing can become repetitive and where automated regression coverage is valuable.

The Selenium project was therefore built around two practical flows. The combined flow checks search, results, negative input handling and a booking smoke path. The end-to-end flow keeps the same browser session alive and follows the richer journey up to the payment page.

3. Objectives and Scope

- Automate the main user journey from launching ixigo to reaching the payment page.

- Validate search-page elements, source and destination selection, and flight-result availability.
- Check important result-page actions such as applying the non-stop filter and sorting by price.
- Cover negative scenarios where the user should not be allowed to continue with invalid search input.
- Use external CSV files for login, route and traveller data so that test data remains separate from test logic.
- Generate evidence through screenshots, logs, pytest HTML report output and Allure result files.

4. Test Environment

Area	Details
Language	Python
Automation tool	Selenium WebDriver
Test runner	pytest
Reporting	pytest-html, Allure result attachments and saved screenshots
Browser	Microsoft Edge, visible execution
Design pattern	Page Object Model
Data source	CSV files for route, login and traveller data
Main command	pytest, using the options configured in pytest.ini

5. Framework Architecture

The framework is intentionally simple. The tests describe the scenario, the page classes perform browser actions, and the utility layer takes care of configuration, data reading, logging and screenshots. This separation matters because ixigo pages change state often; keeping selectors and actions in page objects makes maintenance easier than writing everything directly inside test methods.

Layer	Files / folders	Responsibility
Tests	tests/	Contains ordered pytest scenarios for combined checks and the full booking journey.
Pages	pages/	Encapsulates browser interactions for search, results, booking, add-ons, payment and login.
Utilities	utils/	Reads config and CSV data, writes logs, and captures screenshots.
Data	data/	Stores flight routes, login mobile number and traveller details.
Reports	reports/	Stores pytest HTML output, screenshots and Allure result files.

6. Test Data Strategy

The suite avoids hard-coding the main input values directly inside the test methods. Routes are read from flight_routes.csv, traveller details come from traveller_data.csv, and the login mobile number is read from login_data.csv. This is a small detail, but it makes the framework feel closer to a real testing project because data can be changed without touching the automation logic.

flight_routes.csv	Source city, source code, destination city, destination code and travel date.
traveller_data.csv	Title, name, nationality, contact number, email, pincode and address.
login_data.csv	Mobile number used to begin the login flow before OTP entry.

7. Execution and Reporting

The pytest configuration already defines the main reporting options. A normal test run writes a self-contained HTML report to reports/report.html, stores screenshots in reports/screenshots, and produces Allure result files in reports/allure-results. The HTML report is useful for quick submission review, while Allure gives a deeper drill-down view with attachments.

Output	Location	Purpose
HTML report	reports/report.html	Readable browser-based execution summary.
Screenshots	reports/screenshots/	Visual evidence captured during important steps and test completion.
Allure results	reports/allure-results/	Raw result data and attachments for Allure report generation.
Logs	logs/automation.log	Execution trace for debugging and review.

8. Selenium Test Case Summary

Result: The Selenium evidence available in the project shows the combined flow and the richer full end-to-end flow. Together they cover 24 checks: 9 combined-flow scenarios and 15 end-to-end booking steps.

No.	Suite	Test case	Expected result	Status
1	Combined	Open ixigo page	Application opens and title contains ixigo	Passed
2	Combined	Page elements are present	From, To and Search controls are visible	Passed

3	Combined	Search BOM to BLR flights	Flight results load and count is greater than zero	Passed
4	Combined	Flight count and book buttons	Flights and visible Book buttons are available	Passed
5	Combined	Apply filter and sort	Non-stop filter and price sorting keep usable results	Passed
6	Combined	Print flight results	Top results are printed to the logger	Passed
7	Combined	Same source and destination	Invalid same-route search does not navigate to results	Passed
8	Combined	Search without destination	Search is blocked when destination is missing	Passed
9	Combined	Booking flow smoke check	Search to booking page is completed	Passed
10	E2E	Open site	Flight search page opens in the same browser session	Passed
11	E2E	Login	Mobile number is submitted and OTP wait is handled	Passed
12	E2E	Search flights	CSV route is searched and results are loaded	Passed
13	E2E	Apply non-stop filter	Filtered results remain available	Passed
14	E2E	Sort by price	Sorted results remain available	Passed
15	E2E	Print results	At least one flight result is available to print	Passed
16	E2E	Click Book	Book action is triggered for the selected flight	Passed
17	E2E	Booking page	Booking URL/page state is detected	Passed
18	E2E	Decline cancellation	Free-cancellation prompt is handled	Passed
19	E2E	Fill traveller	Traveller form is filled from CSV data	Passed
20	E2E	Continue and confirm	Traveller details are confirmed	Passed
21	E2E	Add-ons page	Add-ons page loads after booking confirmation	Passed
22	E2E	Select seat	Seat is selected when available, otherwise flow continues	Passed
23	E2E	Skip to payment	Add-ons are skipped and payment navigation begins	Passed
24	E2E	Payment page	Payment page is reached successfully	Passed

9. Screenshot Evidence

The following screenshots are taken from the Selenium execution and are placed in the same order as the user journey. They show the automation moving through the major checkpoints rather than only listing the final pass status.

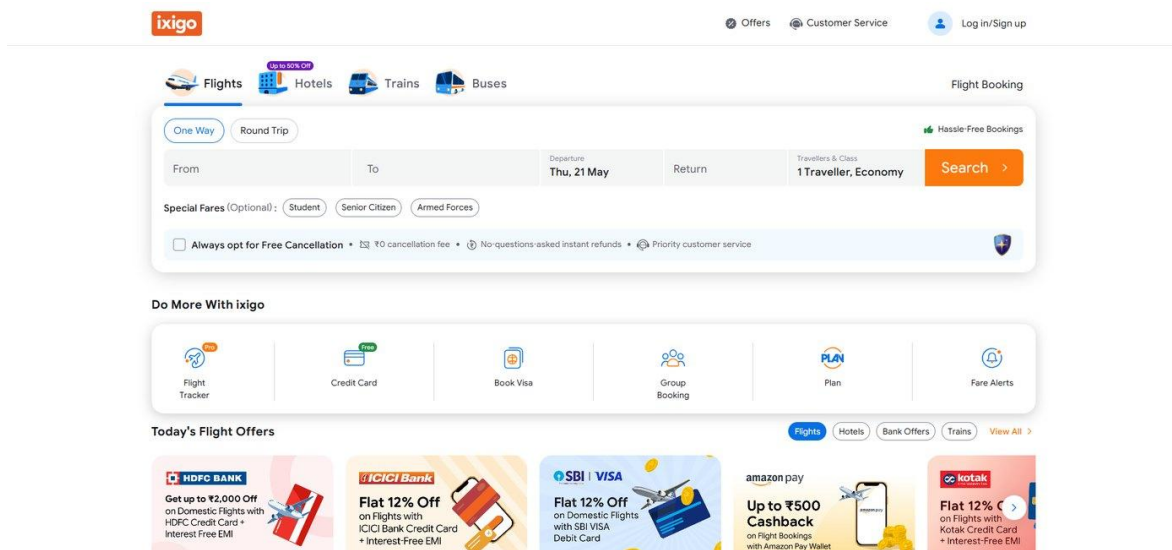


Figure 1. ixigo flight search page opened successfully.

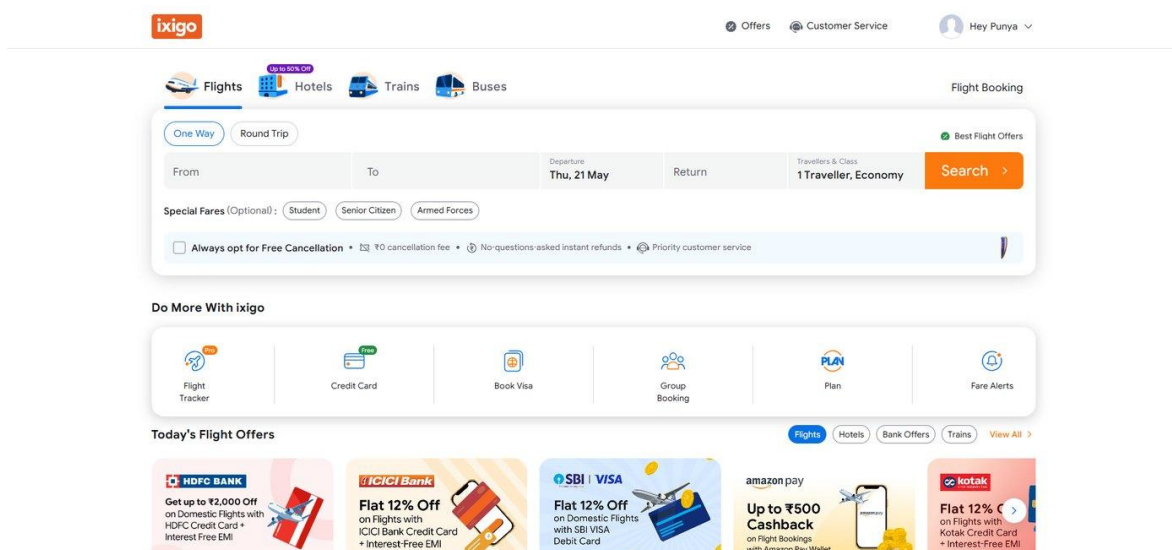


Figure 2. Login flow started with the mobile number from CSV test data.

ixigo Offers Customer Service Hey Punya

Flights Hotels Trains Buses Flight Booking

One Way Round Trip

From: BOM - Mumbai To: BLR - Bengaluru Departure: Wed, 10 Jun Return: Travellers & Class: 1 Traveller, Economy Search

Special Fares (Optional): Student Senior Citizen Armed Forces

☐ Always opt for Free Cancellation • ₹0 cancellation fee • No questions asked instant refunds • Priority customer service

Do More With ixigo

Flight Tracker Credit Card Book Visa Group Booking Plan Fare Alerts

Today's Flight Offers

HDFC BANK Get up to ₹2,000 Off on Domestic Flights with HDFC Credit Card + Interest Free EMI

ICICI Bank Flat 12% Off on Flights with ICICI Bank Credit Card + Interest-Free EMI

SBI | VISA Flat 12% Off on Domestic Flights with SBI VISA Debit Card

amazon pay Up to ₹500 Cashback on Flight Bookings with Amazon Pay Wallet

kotak Flat 12% Off on Flights with Kotak Credit Card + Interest-Free EMI

Figure 3. Search form completed before navigation to flight results.

ixigo Offers Customer Service Hey Punya

Flights Hotels Trains Buses More

From: BOM - Mumbai To: BLR - Bengaluru Departure: Wed, 10 Jun Return: Travellers & Class: 1 Traveller, Economy Search

Special Fares (Optional): Student Senior Citizen Armed Forces

Filters Non-Stop

Smart Filters

Ask something like: I want to see flights with no stops under ₹5000.

Filter Flights

Recommended Filters

Non-Stop ☒ Free meal available ☐ Air India ☐ IndiGo ☐

Stops

Non-Stop ₹5,308 1 Stop ₹3,999 2+ Stops

Up to ₹2,000 Off with HDFC Bank Credit Card EMI Flat 12% Off with Kotak Retail Credit Card EMI Flat 12% Off with SBI VISA Debit Cards

36 Flights Available

Sort by Price Low to High Fastest Shortest First Departure Earliest First Smart Recommended

SpiceJet SG669 19:50 BOM 2h Non-stop 21:50 BLR ₹5,566 Extra ₹834 Off Book Lock Price @ ₹939 Flight Details

SpiceJet SG559 22:30 BOM 2h Non-stop 00:30⁺ BLR ₹5,566 Extra ₹834 Off Book Lock Price @ ₹649 Flight Details

Getting all the available flights

Akasa Air 23:20 3h 50m 01:10⁺ ₹5,308 Book

Figure 4. Non-stop filter applied on the results page.

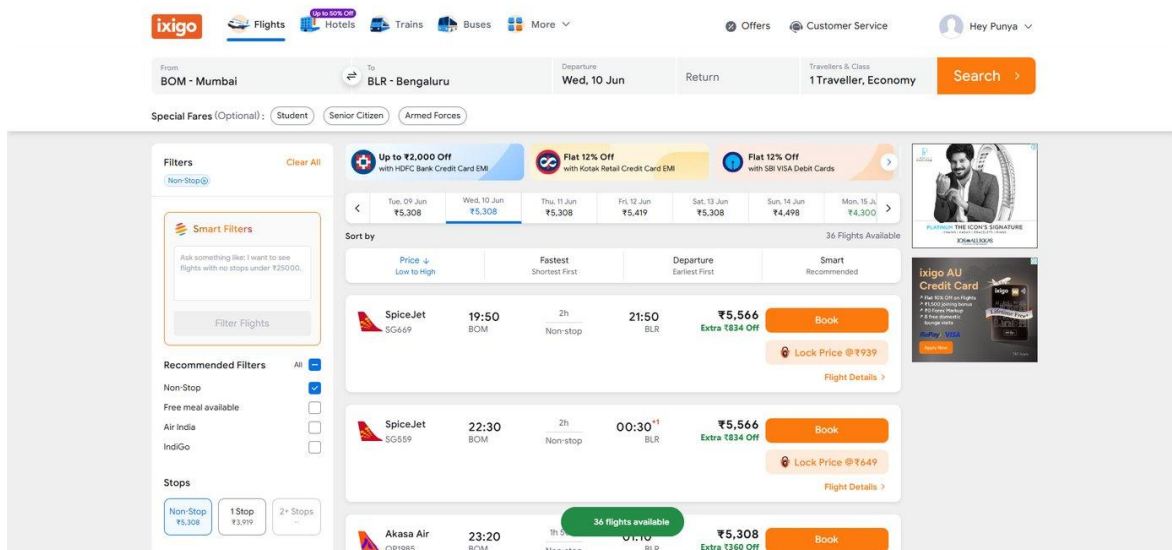


Figure 5. Results sorted by price after filtering.

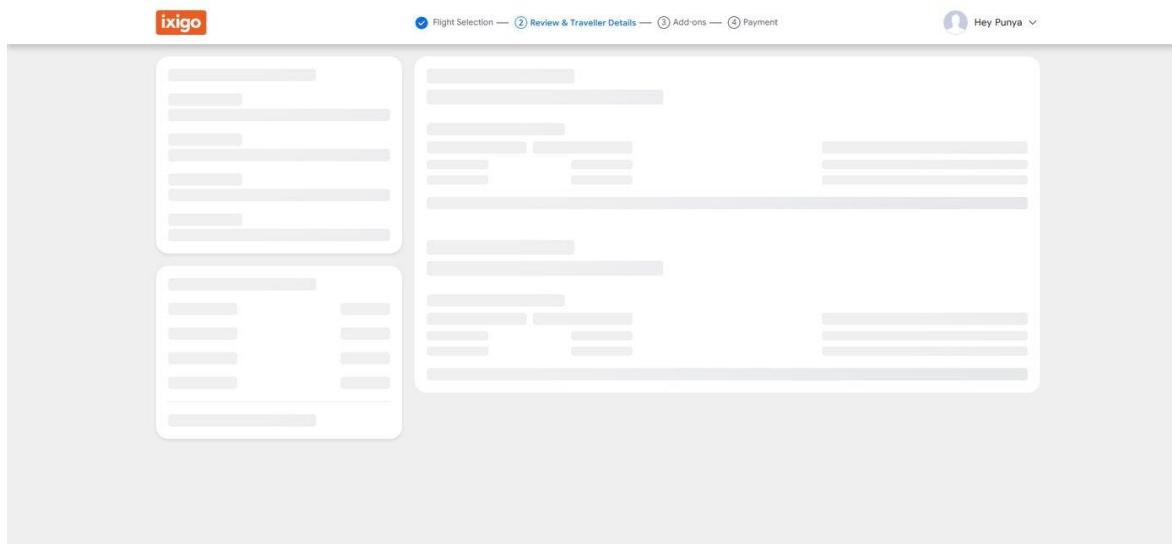


Figure 6. Cheapest available flight selected for booking.

ixigo Flight Selection — 2 Review & Traveller Details — 3 Add-ons — 4 Payment Hey Punya

Offers For You

Have a promo code? Redeem here.

- SALE** ₹834 Off
Coupon applied! Enjoy an instant discount of ₹834. [Know More](#)
- IXRBLEMI** ₹667 Off
Flat 12% Off up to ₹1,500 with RBL Bank Credit Card + Interest Free EMI. [Know More](#)
- SBVISADCD** ₹667 Off
Flat 12% Off up to ₹1,500 with SBI VISA Debit Card. [Know More](#)

[View All Offers >](#)

Fare Summary 1 Traveller

Fare Type	Partially Refundable
Base Fare	₹4,200
Taxes & Fees	₹1,366
Instant Off	-₹834
Total Amount	₹4,732

Mumbai → Bengaluru
Wed, 10 Jun • Non-stop • 2h • Economy

SpiceJet 15G 669

Wed, 10 Jun
19:50
BOM - Mumbai
Mumbai Chhatrapati Shivaji International Airport
Terminal 1

2h

Wed, 10 Jun
21:50
BLR - Bengaluru
Kempegowda International Airport
Terminal 1

Baggage
Cabin : 7 kg per adult
Check-in : 15 kg per adult

Refund on Cancellation [Cancellation & Rescheduling Policy >](#)

BOM - BLR

₹533 refund

Now 19:50 06 Jun 19:50 Departure 10 Jun 19:50

Traveller Details
Choose from the saved list or add a new passenger Traveller

Please ensure that your name matches your govt. ID such as Aadhaar, Passport or Driver's License

₹4,732 ~~₹5,566~~ [Lock Price @₹1939](#) [Continue >](#)

Figure 7. Free-cancellation prompt handled during booking.

ixigo Flight Selection — 2 Review & Traveller Details — 3 Add-ons — 4 Payment Hey Punya

Fare Summary 1 Traveller

Fare Type	Partially Refundable
Base Fare	₹4,200
Taxes & Fees	₹1,366
Instant Off	-₹834
Total Amount	₹4,732

GST Details
To claim credit of GST charged by airlines, please enter your GST details.

☐ I would like to add my GST Number.

Billing Address
As per the latest govt. regulations, it's mandatory to provide your address.

Pincode: 560001 Address: 42 MG Road Bengaluru
City: Bengaluru State: Karnataka

By clicking on continue, I confirm that I have read, understood, and agree with the [Fare Rules](#), [Privacy Policy](#) and [Terms of Use](#).

100% Safe Payment Process

[Verified by Visa](#) [SafeKey](#) [MasterCard SecureCode](#)

₹4,732 ~~₹5,566~~ [Lock Price @₹1939](#) [Continue >](#)

Figure 8. Traveller details filled from the CSV data file.

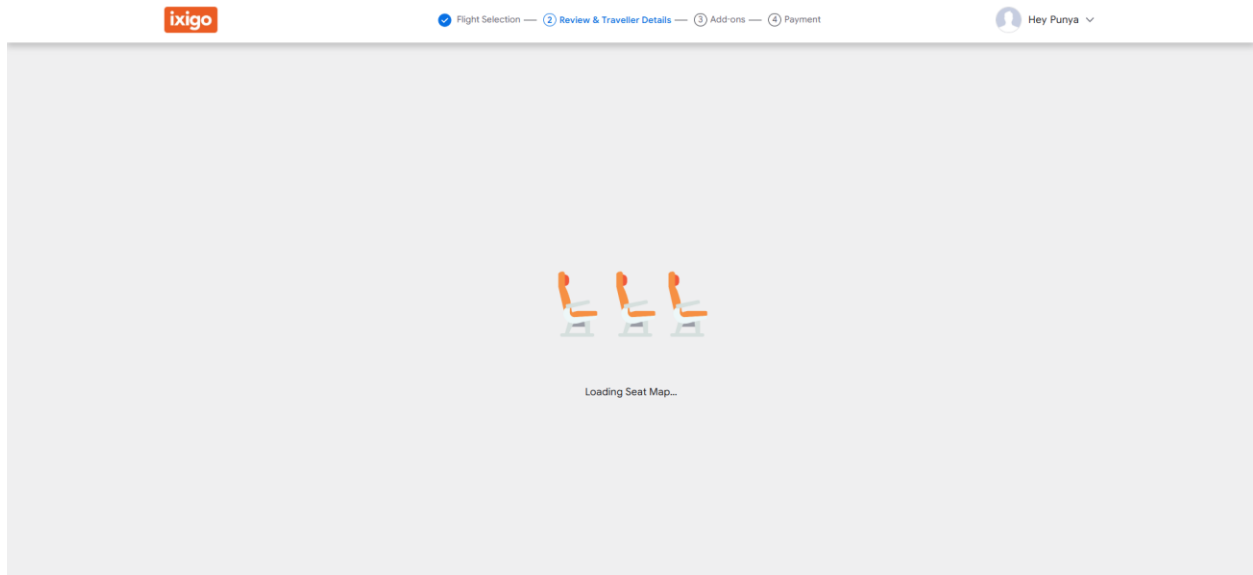


Figure 9. Add-ons page reached after traveller confirmation

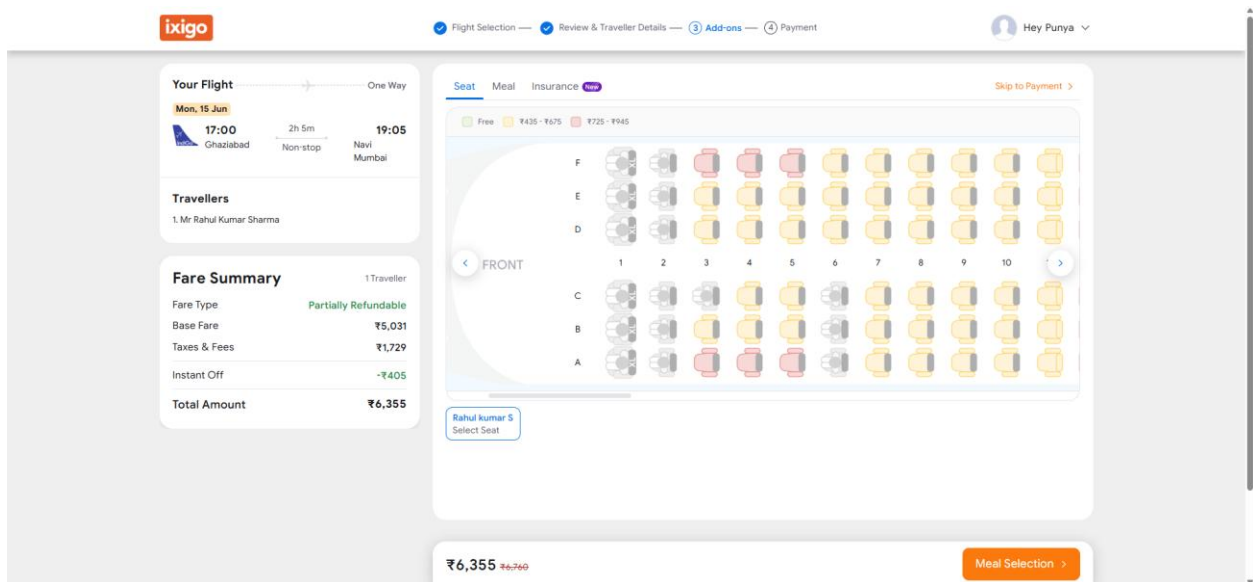


Figure 10. Seat selection step completed or skipped depending on availability.

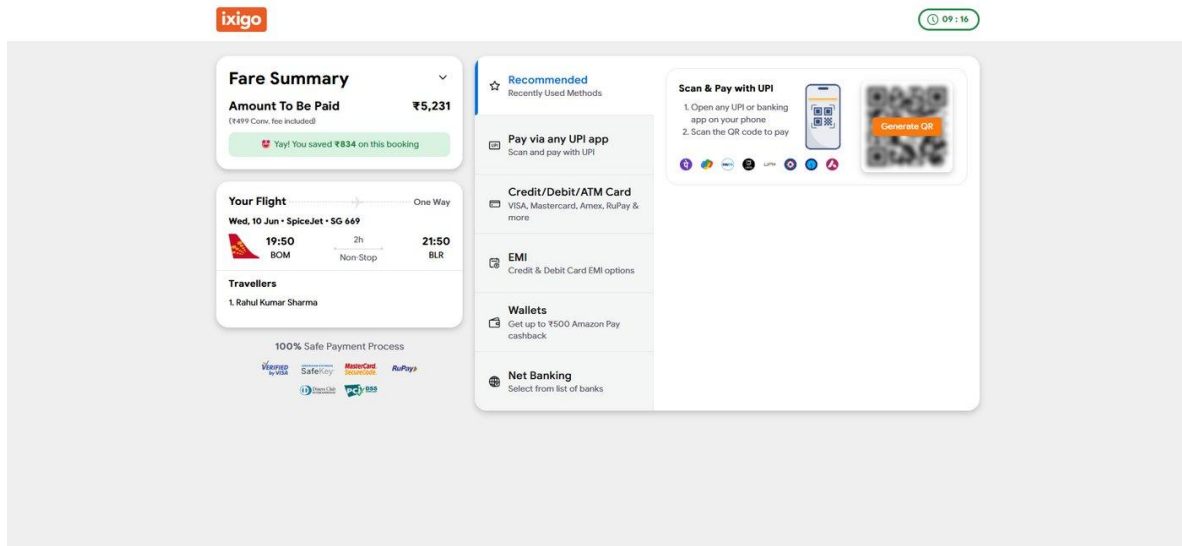


Figure 11. Payment page reached, completing the end-to-end flow.

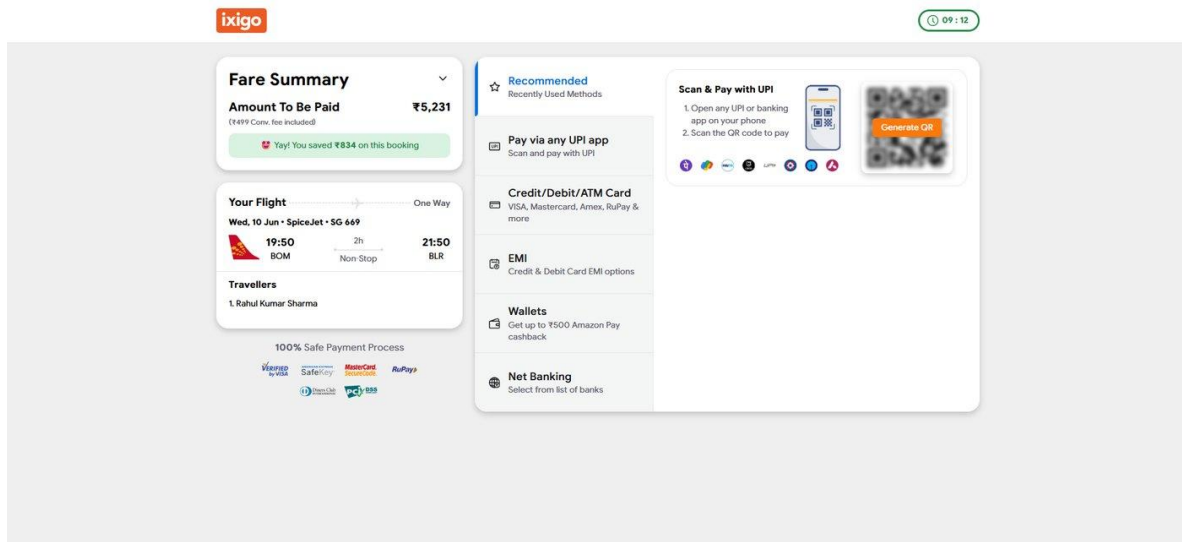


Figure 12. Final captured state after the full booking journey.

10. Observations

- The Page Object Model made the flow easier to read because test methods stayed close to business behaviour.
- The richer end-to-end file is useful for demonstration because the browser session remains continuous from login to payment.
- Negative tests add balance to the suite by proving that invalid input does not quietly move to the results page.
- Screenshots are captured at meaningful points, which makes the report easier to verify without replaying the full run.

11. Conclusion

The Selenium automation project successfully covers the important ixigo flight booking path and presents the result in a way that can be reviewed by both a technical evaluator and a project guide. The framework uses a sensible structure, external test data, browser-level validation, screenshots and report output. Overall, the project demonstrates practical web automation rather than a collection of isolated click scripts.